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Notes

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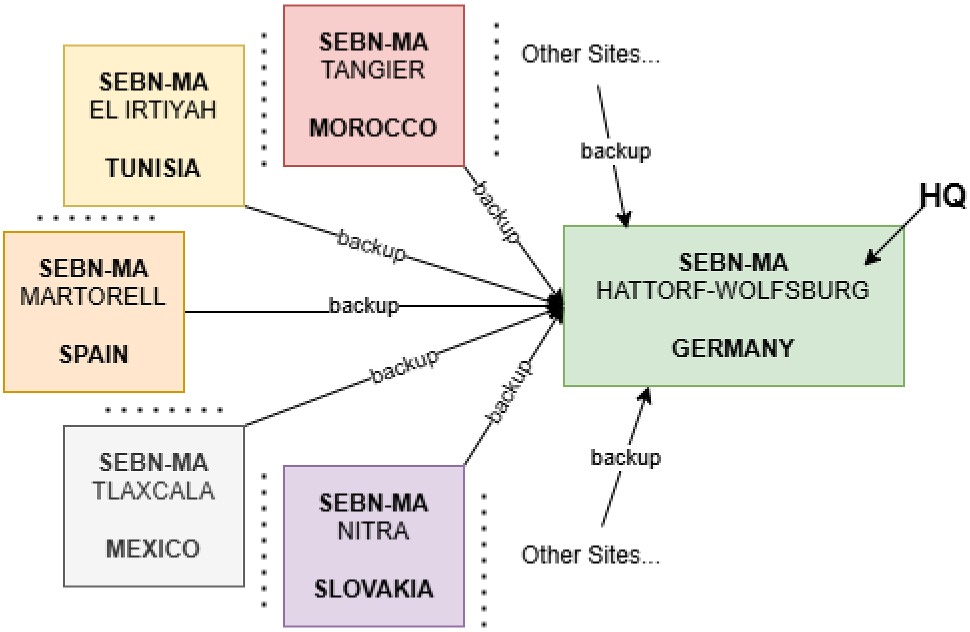
Process Handling at SEBN-MA:

Backup Process:

- Made everyday at night **automatically**, without any interaction by the user.
- 1 specific team in the SEBN Headquarters (in Hattorf-Wolfsburg, Germany) handles all of the requests of backuping/restoring the data (system files, configurations, logs, etc) for any material inside the company (switch, router, etc).



- Previously a robot used to handle all of these requests, but it became out of service because it had become obsolete.

Problem Diagnostic Process:

- Log files are used primarily to diagnose a problem (but sometimes a simple problem can be diagnosed with just the experimented naked eyes).
- Log files contain the time and date of the exact operation, each of those operations is guaranteed to be written in the log system, and then it can be exported for further use or investigation.

N.B.: Each system is connected to a UPS (Uninterrupted Power Supply), so that when losing the power, the system has enough time to rewrite those log files from its internal RAM cache back into the hard drives.

- The process of gathering log files can be handled through two different methods:
    - **Manually:** by typing *sh log* or *show logging* in the systems terminal.

- **Semi-Automatic:** by mimicking the manual typing of the command, but for a lot more systems, a program (logElastic for example) can handle this efficiently.
- **Automatically:** Through a portal named the IPAM (IP Address Management), that will absolutely remove all of the hassle of manual typing. Though it can become a bit pricey for enterprise use.

Problem Solving Process:

- Made by the helpdesk team composed of four members and a team leader (ELBAQALI TAHIRI Rachid), and done using a ticketing system for transparency and trackability.
- The ticketing system (SysAid) is used to manage all of the IT problems in the help desk (which is named the first point of contact for any IT/Network problem in the enterprise), and then after registering the ticket, it gets delivered to the right department and service (some profound problems and more difficult get forwarded directly to the HQ where the central IT department is). Like most programs around the globe that are used inside enterprises, it can get pricey for its licence to be paid, but it's very efficient.
- For far away tasks, administrators and technicians have remote access to the machines through a remote control app (UltraVNC/CM Remote Control Viewer) that enables them to troubleshoot and solve the problem wirelessly and easily.

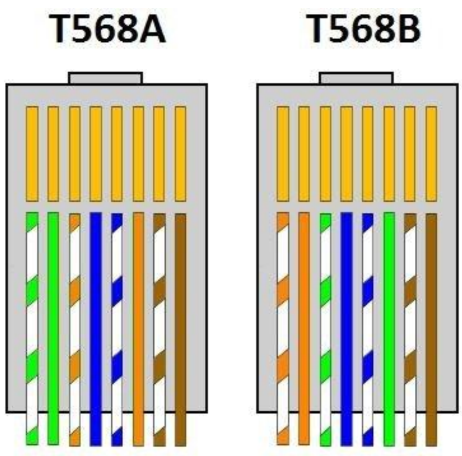
Software Deployment:

- The helpdesk within the IT department uses a PXE (Preboot Execution Environment) that enables them to install predefined OS images into multiple systems at the same time. The PXE also houses multiple versions of predefined programs and softwares that are there to use, so the IT helpdesk doesn't need to download it each time it needs it. The PXE also can replace the HDDs on the machines, so that a machine can work entirely without an HDD, but on the PXE

N.B. : The device needs to be PXE enabled and support it for this to work..

- The company uses its own internal store called "Software Deployment" that handles basic and specialized software since all of the pc programs require an administrator password to be installed. And so to prevent people from coming all day requesting the password just to install basic software, this helps prevent this.

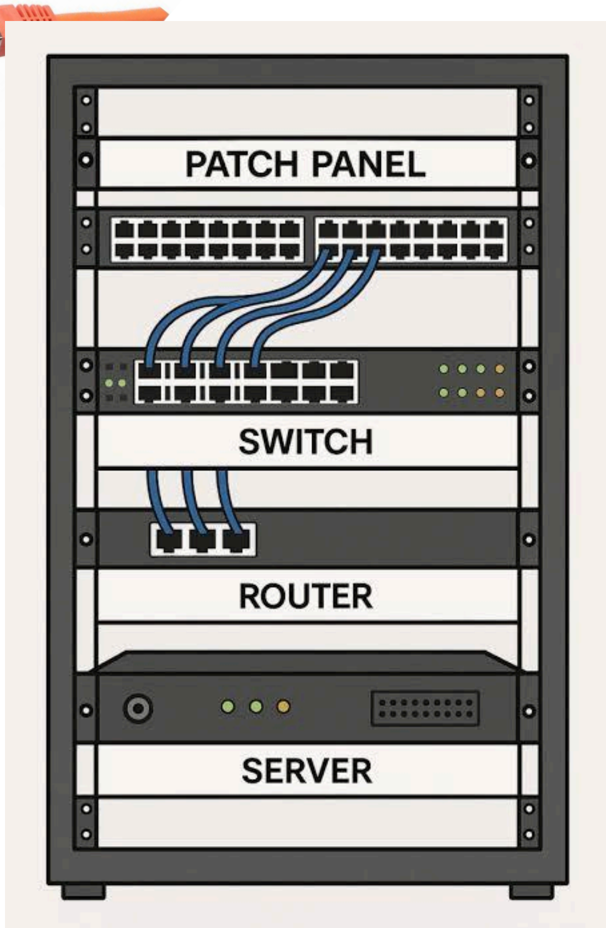
<u>RJ45 Manual Manufacturing Process:</u>



- There are two main types of RJ45 assembly types, type A and B:
- SEBN-MA uses a **type B** as its main way of assembling RJ45 cables, which is a straightforward process.
- These ethernet cables are used to connect machines to the socket up top of the machine , which goes next to the server armoire, which has a patch panel that organizes all of the ethernet cables, and then connects them to the switch.









- **Fun fact:** You can do any combination of RJ45 inner cables layout, but <u>under the condition</u> that the other side of the connector also needs to be in the same layout, since modern computers have come to a way that they can read any combination whatsoever with no problem; But for an external technician for example, it'll be chaos to see, monitor and troubleshoot, which brings us to the layout type B that SEBN-MA uses, so that everything is unified.

General Production Process Overview:

- The general process at SEBN-MA starts with the following, the client submits a request of an order, and then the order gets right into the ERP (Enterprise Resource Planning), where then it will get send to the logistics department for it to handle all of the necessary production materials and then order them (all of the modules, elements, etc), and then after getting them ready in the ERP, since each module inside the company is related and connected to the ERP in some way, and then the necessary materials / modules are ordered or seen if they are available in stock, and then they get imported to the logistics for them to be used and put there.
- The company production process is divided into three main parts:
    a. **CAO (Cutting Area Operations) :** Here is the first step, the cable gets cut into multiple lengths depending on the clients order, they first implement an echantillon, and based on it they send it to the client and see whether it meets the requirements or not, if yes, they continue forward with the production, unless there is some adjustments to be made.
    b. **WAO (Welding Area Operations) :** Here the cable that has been cut, get's taken here for the connector to be put in it and welded in it.
    c. **MAO (Manufacturing Area Operations) :** Here is the last step of the manufacturing of a cable process, it gets divided into a lot of segments but the main thing is that the cable here is either manufactured completely for a car (full wiring harness or 'KSK'), or a small part of that wiring harness that gets undergone under a system called a PLS under the ERP that tells the operator what to do.

N.B. : Before each step, the cables get moved into a temporary TW area where then it gets stored to get moved into the next operation.

- After this global process, it gets taken to be tested (whether electrically, optically, water resistance, air pressure, etc) to meet the clients requirements, and then gets put inside the box and moved into the export area, where there an operator checks all the checkmarks before exporting the order, where when exported and received by the client, the invoice is also sent where then the paiement and financial department handles the rest.

N.Bs. :

- Some very specific or new orders require a prototype to be made and then sent to the client and then for him to confirm commencing the order, which the prototyping area comes into hand, which it simulates all of the processes above, but manually.

- The quality department occasionally comes and checkers for basic damages or specific measures and ensures the process of manufacturing that cable goes and went smoothly
- Each operator in a machine has a magasinier that sees the operation and goes to the logistics or storage compartments and grabs the raw material needed for the order.

The ERP is the one responsible for reuniting all of the programs and processes here, and the ERP has a centralized order portal where it can see all of its orders.